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Awakening sleep nodes: Slack feature enhancement for robust long-term traffic forecasting

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ABSTRACT

Traffic forecasting is crucial for Intelligent Transportation Systems (ITS), especially for enabling accurate long-term and extremely long-term predictions, which are essential for strategic planning and large-scale logistics management. However, achieving robust long-term forecasting remains challenging due to the inherent spatio-temporal aggregation in traffic flow, resulting in many nodes exhibiting consistently low or zero activity—termed sleep nodes. These nodes provide minimal gradient signals during training, hindering global traffic pattern learning and degrading generalization performance. To alleviate this problem, we propose a simple yet effective plug-and-play information fusion method, termed *Slack Feature Enhancement* (SFE). Specifically, SFE computes a Slack Level for each node, inversely quantifying its historical average activity, and fuses this prior knowledge with real-time inputs at the network's input layer. This approach mitigates the lack of informative signals from sleep nodes, improving gradient propagation and enhancing feature discriminability. Extensive experiments on seven real-world datasets—including tasks ranging from short-term to extremely long-term forecasting—demonstrate that SFE significantly boosts forecasting performance. Incorporating SFE notably reduces training time to approximately 71 % of state-of-the-art models' original duration and achieves substantial improvement in extremely long-term forecasting, reducing Mean Absolute Percentage Error (MAPE) by up to 18.64 %. Codes are available at <https://github.com/Hairui-Sun/SFE-Traffic>.

1. Introduction

1.1. Research background

The evolution of urban transportation systems has elevated traffic flow forecasting to a pivotal research domain, emerging as a cornerstone for smart city development, mobility optimization, and sustainable urban planning. Modern sensing infrastructures, deployed extensively across road networks (Fig. 1), generate multivariate time series data that capture spatiotemporal traffic dynamics (Fig. 2). These high-resolution datasets enable systematic analysis of complex traffic patterns through temporal and spatial dimensions, providing unprecedented opportunities for extracting latent transportation dynamics and advancing predictive accuracy through cutting-edge computational approaches [1]. However, while short-term forecasting has seen substantial improvements, reliably predicting long-term and extremely long-term traffic trends remains considerably more challenging due to error accumulation, information sparsity, and dynamic city conditions.

Before the advent of deep learning, traffic forecasting largely relied on statistical models and shallow machine learning techniques. Early approaches such as AutoRegressive Integrated Moving Average (ARIMA) and Support Vector Regression (SVR) [2] offered limited predictive accuracy and struggled to capture the nonlinear, high-order spatio-temporal dependencies inherent in urban traffic. Moreover, they required manual feature engineering and often assumed stationarity or regularity, making them poorly suited for dynamic and heterogeneous city-wide conditions. These shortcomings—particularly their inability to *fuse heterogeneous information over extended horizons*—led to a growing demand for more expressive, data-driven models capable of adapting to complex real-world traffic patterns and delivering reliable long-term forecasts.

With the advancement of deep learning, significant progress has been achieved in traffic flow prediction. Early approaches primarily relied on recurrent architectures such as Long Short-Term Memory (LSTM) [4] and Gated Recurrent Unit (GRU) to capture temporal dynamics [5], while subsequent research introduced graph convolutional

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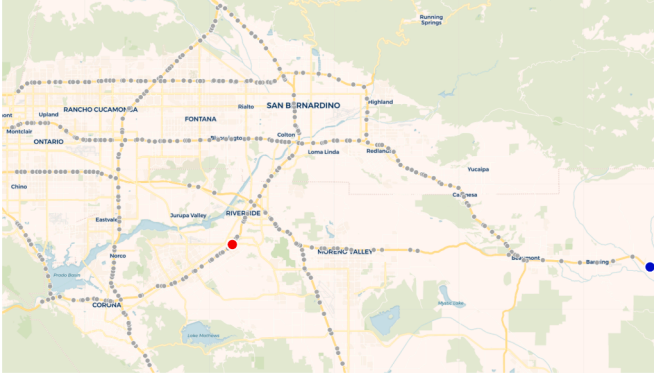


Fig. 1. Sensor deployment on the roads for the PEMS08 dataset (data obtained from the Caltrans Performance Measurement System [3]). Red markers indicate the most active (high-traffic) sensors, blue markers represent dormant nodes with consistently low traffic, and gray markers denote all other sensor locations.

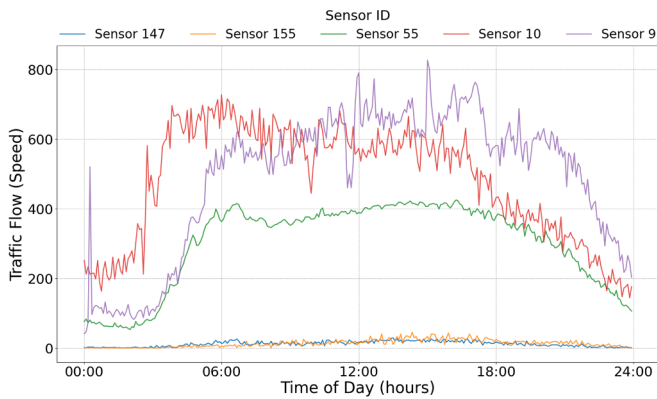


Fig. 2. PEMS08 traffic flow time series for selected sensors. Sensors 9 (red markers in Fig. 1) exhibit high traffic activity and pronounced temporal variations throughout the day, while sensors 147 (blue markers in Fig. 1) maintain consistently low or near-zero traffic flow, exemplifying typical “sleep nodes”.

networks(GCNs) to model spatial dependencies. More recently, attention mechanisms have been incorporated to dynamically weigh spatio-temporal features, resulting in notable performance improvements [6].

1.2. Problem statement

However, despite these advances, most prior works have predominantly focused on short-term predictions [7], which are insufficient for many real-world applications. Only recently, works such as SSTBAN [8] and Extralonger [9] have pushed the field toward long-term and extremely long-term forecasting tasks, where modeling stability and generalization over extended horizons becomes increasingly critical. As deep models are tasked with longer-range prediction, a longstanding yet underexplored challenge becomes especially prominent: a substantial number of nodes in the network persistently exhibit low or near-zero traffic flow, leading to what we term the *sleep node* phenomenon.

Due to the inherent aggregation characteristics of traffic flow, a natural dichotomy emerges: while a subset of nodes exhibits high activity, many others remain dormant. As illustrated in Fig. 2, some sensors display pronounced temporal variations throughout the day, whereas others maintain consistently low or near-constant values. For example, sensor 154 (blue marker in Fig. 1) remain persistently inactive, in contrast to sensor 9 (red marker), which exhibit substantial fluctuations. Their spatial distribution is visualized in Fig. 1, highlighting the contrast between active and dormant regions. After normalization—especially in datasets containing both types of nodes—dormant nodes tend to suffer extreme compression of signal values. **This makes it difficult for the**

model to learn from these nodes and can lead to a biased representation of the overall traffic network, severely degrading its ability to capture global spatio-temporal dependencies, particularly under long forecasting horizons.

1.3. Proposed solution overview

To mitigate these challenges, we propose a novel input-level feature augmentation strategy: **Slack Feature Enhancement (SFE)**. The core idea is to pre-process historical traffic data to compute a node-specific *slack level* s_i , quantifying each node’s intrinsic activity or dormancy over an extended period. This slack level is then fused with real-time traffic measurements, yielding an augmented input representation:

$$\tilde{x}_i = [x_i, s_i],$$

where x_i denotes the real-time traffic features. **This enhanced representation provides two key benefits:** (1) it supplements the sparse informative signals from dormant nodes by providing a stable, context-rich feature, and (2) it enhances the feature discriminability between active and dormant nodes, guiding the model to better capture global structures.

The overall SFE process is visualized in Fig. 3.

1.4. Paper organization

This paper is organized according to the following research roadmap:

- **Problem Background and Motivation:** We preliminarily identify the sleep node phenomenon as a key challenge in traffic forecasting (Section 1), and further investigate prior research to confirm that this issue has been largely overlooked in existing studies (Section 2).
- **Methodology Design and Theoretical Foundations:** We first introduce feature fusion as a common strategy to augment model inputs in scenarios where signals are weak or sparse, highlighting its suitability for addressing the sleep node phenomenon (Section 3). We then comprehensively describe the proposed Slack Feature Enhancement (SFE) structure and integration methods (Section 4.1). Furthermore, we conduct an in-depth theoretical analysis (Section 4.2) to answer three critical questions: (1) whether dormant nodes indeed impair model learning; (2) whether feature fusion effectively mitigates this problem; and (3) whether the specific design choices of SFE are theoretically justified.
- **Experimental Validation and Analysis:** To systematically bridge motivation, methodology, and evaluation, we formulate nine concrete research questions (Q1-Q9) that guide our experimental section (Section 5). Each question is explicitly mapped to a core challenge, theoretical insight, or methodological objective highlighted earlier in the paper. All experiments—including ablation studies, robustness tests, and state-of-the-art comparisons—are designed to comprehensively address these questions, ensuring a coherent flow from scientific motivation to empirical validation.

1.5. Contributions

This work makes the following key contributions:

- We identify and formalize the *sleep node* phenomenon in traffic forecasting, highlighting its detrimental impact on long-term prediction accuracy through theoretical analysis and empirical evidence.
- We propose **Slack Feature Enhancement (SFE)**, a lightweight, plug-and-play feature-fusion method that effectively mitigates the sleep node issue and improves model robustness without requiring architectural changes.
- We conduct extensive experiments and theoretical studies demonstrating that SFE substantially improves both long-term and extremely long-term traffic forecasting accuracy, achieving state-of-the-art results across diverse benchmarks while reducing training time.

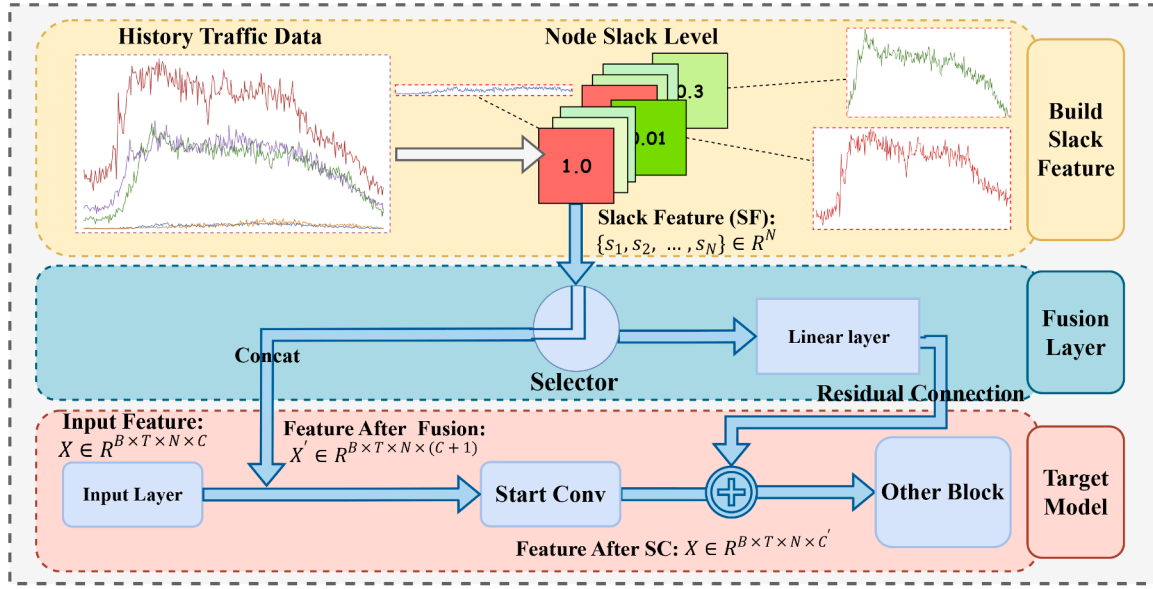


Fig. 3. Overview of the Slack Feature Enhancement (SFE) process. Historical traffic data (yellow area) are used to compute node-specific Slack Levels. These are fused with real-time traffic inputs via concatenation or residual connections (green area), producing an augmented representation for the forecasting model (pink area).

2. Related work

In this section, we provide a comprehensive review of the literature on traffic forecasting. Our analysis charts the field's trajectory from early statistical models to the current era of complex spatio-temporal deep neural networks. We will demonstrate that this evolution has been predominantly **model-centric**, focusing on designing increasingly sophisticated architectures. This consistent focus, however, has led to a critical oversight of fundamental, **data-centric** challenges, chief among them the **sleep node problem** that is the focus of this paper.

2.1. Early statistical and machine learning approaches

The foundational challenge of traffic forecasting—predicting future traffic states from historical time-series data—was initially addressed by traditional statistical methods. Models such as Historical Average (HA) [10] and the Autoregressive Integrated Moving Average (ARIMA) family [11] were widely adopted due to their interpretability and straightforward implementation. Alongside these, state-space models like the Kalman Filter (KF) were also employed. However, these methods are fundamentally constrained by their underlying assumptions of linearity and stationarity, which limits their ability to capture the highly non-linear and dynamic nature of real-world traffic patterns.

To better handle non-linearity, various machine learning techniques were explored. These methods often relied on sophisticated feature engineering to achieve their goals. For example, Dell'Acqua et al. introduced a time-aware multivariate nearest neighbors regression model [12], while Luo et al. developed a hybrid method that combined Support Vector Regression (SVR) with the discrete Fourier transform to model periodicity [13]. While more powerful than statistical models, their performance was heavily dependent on the quality of hand-crafted features, demanding significant domain expertise and limiting their end-to-end learning capability.

2.2. The evolution of spatio-temporal deep learning

The advent of deep learning revolutionized the field by enabling automatic feature extraction and end-to-end model training, leading to significant performance breakthroughs.

2.2.1. From temporal modeling to spatial grids

Early deep learning applications focused primarily on capturing temporal dependencies. Recurrent Neural Network (RNN) variants, such as Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU), became standard tools for sequence modeling in traffic forecasting [5]. Soon after, researchers began to explicitly model spatial correlations. A common initial approach was to represent road networks as Euclidean grids and apply Convolutional Neural Networks (CNNs). A seminal work in this area is ST-ResNet [1], which utilized a residual CNN framework to model spatial proximity in traffic data, demonstrating a remarkable 26.7% RMSE reduction over ARIMA on the TaxiBJ dataset. Other works also explored CNN-based frameworks, for instance, by combining them with clustering methods [14]. The primary limitation of this entire line of research, however, is the inherent inability of CNNs to effectively model the non-Euclidean topology of real-world road networks.

2.2.2. The paradigm shift to graph neural networks

To address the topological limitations of CNNs, the field underwent a paradigm shift towards Graph Neural Networks (GNNs), which naturally represent road networks as graphs. The versatility of GNNs has been proven across numerous domains beyond traffic, such as social computing [15], emotion recognition [16], and graph-based federated learning [17]. Foundational models like T-GCN [18] and STGCN [19] were pioneers, effectively combining graph convolutions with temporal modules to learn unified spatio-temporal representations. STGCN, for instance, lowered the 15-minute MAE on PEMS-BAY by 8.2% compared to T-GCN, establishing a strong benchmark.

This led to an explosion of research focused on enhancing the graph learning component. One major thrust was improving the expressiveness of the graph structure itself. STMGCN, for example, proposed a multi-graph approach to capture different types of spatial relationships simultaneously [20]. A critical innovation came with GWNET [21], which introduced a self-learning adaptive adjacency matrix, freeing models from reliance on predefined, often suboptimal, static graphs. This advancement further improved performance, with GWNET achieving an 11% gain over STGCN on the PEMS-BAY dataset. The pursuit of capturing time-varying dynamics led to a new class of dynamic graph models. These include DMSTGCN [22], DSTAGNN [23], TPGCN [24], and PGCN

[25], each proposing unique mechanisms to allow the graph structure to evolve over time.

2.2.3. The role of attention mechanisms and long-term forecasting

Another significant line of research involves the integration of the attention mechanism [26] to dynamically weigh the importance of different spatial and temporal features. This has led to a variety of attention-based spatio-temporal GNNs, including ASTGCN [27], and MISTAGCN [28]. Some methods also combine attention with other techniques, such as Fourier series to model periodicity [29], or use attention to adjust static weights [30]. This architectural sophistication has been pushed to its limits in the particularly challenging sub-field of long-term forecasting. Models like SSTBAN [8] and, more recently, Extralonger [9] have introduced specialized long-dependency modeling techniques to extend prediction horizons to a week or more. Other research has focused on specialized tasks [31], such as incorporating uncertainty quantification with STUP [32] or learning from pivotal graphs with STPGNN [33].

2.3. Identifying the research gap: A data-centric perspective

This extensive body of literature, from ARIMA to cutting-edge models like Extralonger, demonstrates a clear and consistent trajectory: designing more powerful and complex **model architectures** to better fit the observed data. The quantitative improvements shown by models like ST-ResNet, STGCN, and GWNET are testaments to the success of this model-centric race. Yet, this entire paradigm hinges on a crucial, unexamined assumption: that the data from every sensor node is consistently meaningful.

While works like HSTGNN [34] focus on **architectural innovation**—introducing a novel end-to-end model with a *decoupling layer* and *hybrid graph learning* to address known challenges—our work is fundamentally different. We introduce SFE, a lightweight, model-agnostic **feature engineering** module, to solve the previously overlooked, data-level problem of sleep nodes.

Our work challenges this assumption. We identify a fundamental, data-level problem—the persistent low-activity or inactivity of certain sensors, which we term ‘**sleep nodes**’. These nodes provide little to no useful information for forecasting and can act as noise that degrades the performance of the very deep learning models designed to learn from them, a problem that is likely exacerbated over longer prediction horizons. Despite the exhaustive research into model design, to the best of our knowledge, no prior work has explicitly addressed the challenge posed by these sleep nodes. This reveals a critical gap: the need for a **data-centric** approach that enhances data quality before the modeling stage. Our proposed method, SFE, is designed to fill this gap by offering a lightweight, model-agnostic solution.

3. Problem formulation and key concepts

This section provides a formal description of the traffic flow prediction task and introduces two essential concepts—feature fusion and residual embedding—that are incorporated in our method.

Traffic flow prediction

Consider a traffic road network represented as a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ denotes the set of N nodes (i.e., monitoring stations) and $\mathcal{E}$ represents the set of edges between nodes, described by an adjacency matrix $\mathbf{A}$.

Historical traffic measurements are denoted as $\mathbf{X} = [\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_{T-1}] \in \mathbb{R}^{T \times N \times C}$, where T is the historical time window length, N is the number of monitoring stations, and C corresponds to the number of features (e.g., $C = 1$ for speed, or $C = 3$ including speed, time of day, and day of the week).

The objective is to learn a predictive model F that estimates future traffic states $\hat{\mathbf{Y}} = [\hat{\mathbf{x}}_T, \dots, \hat{\mathbf{x}}_{T+T'-1}]$, where T' is the forecasting horizon.

Formally:

$$\mathbf{X} = [\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_{T-1}] \xrightarrow{F(\cdot|\Theta)} \hat{\mathbf{Y}} = [\hat{\mathbf{x}}_T, \hat{\mathbf{x}}_{T+1}, \dots, \hat{\mathbf{x}}_{T+T'-1}] \quad (1)$$

where Θ represents the parameters of $F(\cdot)$, optimized to minimize the discrepancy between the predicted values $\hat{\mathbf{Y}}$ and the ground truth $\mathbf{Y}$.

Feature fusion

Feature fusion refers to the process of combining multiple feature sources into a unified representation that enriches the input to the model. By aggregating complementary information—for instance, temporal, spatial, or contextual features—the model is able to extract more informative patterns and improve prediction accuracy, especially when dealing with heterogeneous or multimodal data inputs.

Residual embedding

Residual embedding integrates auxiliary features into the model via residual connections, inspired by ResNet-style architectures. This approach incorporates additional feature information while preserving the original input. Formally, given main input data $\mathbf{X} \in \mathbb{R}^{T \times N \times C}$ and auxiliary features $\mathbf{Z} \in \mathbb{R}^{T \times N \times D}$, the fused representation $\mathbf{F}$ is computed as:

$$\mathbf{F} = \mathbf{X} + \text{Embed}(\mathbf{Z}) \quad (2)$$

Here, $\text{Embed}(\mathbf{Z})$ denotes a transformation function that aligns the dimension of $\mathbf{Z}$ with that of $\mathbf{X}$. This mechanism enhances the model's expressive power by capturing richer feature interactions while maintaining the consistency of the original input signal.

4. Methodology

This section elaborates our approach in two parts. [Section 4.1](#) details the *Slack Feature Construction Pipeline*, covering how slack information is generated from historical data and how it is embedded into a forecasting backbone. [Section 4.2](#) provides a theoretical discussion centered around critical challenges in long-horizon traffic forecasting, and systematically analyzes how the proposed Slack Feature Enhancement (SFE) addresses these issues.

4.1. Slack feature construction pipeline

Slack feature generation

Given historical traffic observations $\mathbf{X} \in \mathbb{R}^{T \times N \times C}$ (T : time steps, N : nodes, C : feature dimensions), we generate a scalar slack score for each node in two steps.

Step 1: Node-level activity.

$$\bar{v}_i = \frac{1}{T} \sum_{t=0}^{T-1} X_{t,i,0}, \quad i = 1, \dots, N. \quad (3)$$

Step 2: Min-max inversion.

$$\text{SlackLevel}_i = \frac{\max_j \bar{v}_j - \bar{v}_i}{\max_j \bar{v}_j - \min_j \bar{v}_j}, \quad i = 1, \dots, N. \quad (4)$$

The resulting vector $\mathbf{m} = (\text{SlackLevel}_1, \dots, \text{SlackLevel}_N)^T \in \mathbb{R}^N$ is computed **offline once during data preprocessing and remains fixed throughout both training and inference**, ensuring reproducible behaviour without runtime overhead.

Slack feature embedding strategies

To integrate the Slack Feature into the model input, we propose two alternative embedding strategies: direct concatenation and high-dimensional residual embedding. **Importantly, these strategies are intended as independent options rather than jointly applied methods.** Below, we describe each approach in detail.

Let the original model input be $\mathbf{x} \in \mathbb{R}^{B \times T \times N \times C}$ (batch size B), and broadcast $\mathbf{m}$ to $\tilde{\mathbf{m}} \in \mathbb{R}^{B \times T \times N \times 1}$.

Direct concatenation.

$$x_{\text{out}} = \text{concat}(x, \tilde{\mathbf{m}}) \in \mathbb{R}^{B \times T \times N \times (C+1)}. \quad (5)$$

High-dimensional residual embedding.

First project $\mathbf{m}$ to a d -dimensional latent space by default:

$$\mathbf{m}' = \text{FC}(\mathbf{m}) \in \mathbb{R}^{N \times d}, \quad \tilde{\mathbf{m}}' \in \mathbb{R}^{B \times T \times N \times d}. \quad (6)$$

Then inject it additively:

$$x_{\text{out}} = x + \tilde{\mathbf{m}}'. \quad (7)$$

Direct concatenation is parameter-free and computationally light, whereas residual embedding introduces richer interactions via an additional $N \times d$ parameter matrix; practitioners may choose either strategy according to model capacity and deployment constraints.

4.2. Theoretical rationale and discussion

In this section, we provide a conceptual discussion regarding the proposed Slack Feature Enhancement (SFE) approach. Acknowledging that a rigorous theoretical proof for the mechanisms of deep learning models is an open challenge, our aim here is not to provide a definitive analysis. Instead, our goal is to discuss the key questions and rationale that underpin the motivation, design, and observed effectiveness of SFE, offering plausible explanations for its behavior. This discussion is organized into three main parts:

1. The Impact of Dormant Nodes on Long-Term Traffic Forecasting:

We first analyze the inherent challenges posed by dormant or low-activity nodes:

- Why Long-Term Zero Traffic Nodes Impair Model Learning (Q1.1)
- Why Sleep Nodes Significantly Affect Extremely Long-Term Predictions (Q1.2)

2. Motivation for Slack Feature Enhancement (SFE):

We then explain why introducing an additional input feature—the Slack Feature—is an effective method to tackle these challenges:

- Why SFE Improves Long- and Extremely Long-Term Forecasting Performance (Q2.1)
- Why SFE Reduces Training Time (Q2.2)

3. Design Choices in Slack Feature Enhancement:

Finally, we justify the specific design decisions involved in constructing and integrating SFE:

- Why is the Slack Feature constructed offline and kept fixed during inference? (Q3.1)
- Why Slack Feature Enhancement Rather Than Architectural Modification? (Q3.2)
- Why Slack Feature Enhancement Is Robust in Practical Scenarios? (Q3.3)
- What Are the Differences Between the Two Embedding Strategies? (Q3.4)

4.2.1. The challenge of sleep nodes: Information loss and biased learning (Q1.1)

The detrimental impact of sleep nodes on model learning is not a simple phenomenon but a multi-faceted challenge rooted in information theory and optimization dynamics. Rather than a single point of failure, sleep nodes introduce several compounding issues, particularly after standard data preprocessing.

Let the raw traffic signal at a high-activity node be denoted by $x_h(t)$ and at a low-activity (sleep) node by $x_s(t)$. By definition, their signal variances differ drastically: $\text{Var}[x_h(t)] \gg \text{Var}[x_s(t)]$. In traffic forecasting, global z-score normalization across all nodes is a common practice:

$$\tilde{x}_i(t) = \frac{x_i(t) - \mu_{\text{all}}}{\sigma_{\text{all}}}, \quad (8)$$

where σ_{all} is dominated by high-activity nodes. This preprocessing step, while necessary, severely compresses the dynamic range of sleep node signals:

$$\text{Var}[\tilde{x}_s(t)] = \frac{\text{Var}[x_s(t)]}{\sigma_{\text{all}}^2} \ll 1. \quad (9)$$

This quantifiable loss of signal variance is the primary source of several critical consequences for the model:

- **Information Loss and Feature Ambiguity:** As shown in Eq. (9), the unique patterns of different sleep nodes are compressed into a numerically indistinguishable, near-zero range. This prevents the model from learning a discriminative representation, effectively treating these nodes as a single, uninformative group and thereby losing crucial local information.
- **Biased Learning Dynamics:** The optimization process is driven by the gradient of the loss function, which is naturally dominated by nodes that contribute most to the overall error. Due to their compressed signal variance, the loss contribution from sleep nodes is minimal. Consequently, the parameter updates are overwhelmingly guided by the high-activity nodes. While the gradients for sleep node parameters (w_s) may not be strictly zero, their influence is negligible compared to that of active node parameters (w_h):

$$\left\| \frac{\partial L}{\partial w_s} \right\| \ll \left\| \frac{\partial L}{\partial w_h} \right\|. \quad (10)$$

This imbalance fosters a biased model that fails to develop a holistic understanding of the entire traffic network.

- **Impaired Global Feature Representation:** A model that is biased towards high-activity nodes cannot effectively learn the complex interplay between different parts of the network. The subtle influence that low-activity local roads might have on major highways is never captured, leading to an incomplete and fragile representation of global spatio-temporal dynamics.

4.2.2. Compounding effects in extremely long-Term predictions (Q1.2)

The challenges posed by sleep nodes are significantly amplified in extremely long-term forecasting scenarios, where the model's stability and understanding of long-range dependencies are paramount. The initial issues of information loss and biased learning do not remain static; they compound over extended prediction horizons through several mechanisms.

1. **Accumulation of Representational Deficits:** In auto-regressive forecasting, the prediction at step t is used as input for step $t+1$. The model's initial inability to accurately represent a sleep node s leads to a prediction error, $\varepsilon_{s,t} = \hat{y}_{s,t} - y_{s,t}$. The input for the next prediction is thus corrupted by this error, as the model computes $\hat{y}_{s,t+1} = f(\hat{y}_{s,t}, \dots) = f(y_{s,t} + \varepsilon_{s,t}, \dots)$. Consequently, the error at each subsequent step becomes a function of all prior errors. Over a long forecasting horizon of length T' , the final error for the sleep node can be expressed conceptually as:

$$\text{Error}_{t+T'} = g(\varepsilon_{s,t}, \varepsilon_{s,t+1}, \dots, \varepsilon_{s,t+T'-1}). \quad (11)$$

This demonstrates how small initial representational deficits can progressively corrupt the future state, causing prediction errors for these nodes to accumulate and potentially grow unboundedly over time.

2. **Failure to Model Long-Range Dependencies:** Long-term traffic patterns are often governed by subtle, long-range causal effects (e.g., a minor blockage on a local road slowly creating a traffic jam on a distant highway hours later). A model that is effectively "blind" to the dynamics of sleep nodes (the local roads) from the outset can never learn these crucial long-range dependencies. It lacks the necessary information to connect cause and effect over extended spatio-temporal distances.

3. Increased Model Instability: A model forced to predict the behavior of parts of a system it does not understand is inherently unstable. Without a robust representation of low-activity nodes, the model may treat their weak signals as random noise, leading to erratic and unreliable predictions over long horizons.

In summary, the seemingly minor issue of low-activity nodes evolves into a critical bottleneck for long-term forecasting. The cumulative effect of information loss and biased learning systematically undermines the model's ability to capture the complete, long-range dynamics of the traffic system, leading to a significant degradation in performance.

4.2.3. Plausible mechanisms for SFE's performance enhancement (Q2.1)

The Slack Feature Enhancement (SFE) method is designed to address the challenges of information loss and biased learning caused by sleep nodes. While the precise mechanisms within a complex deep learning model are difficult to isolate and prove rigorously, we hypothesize that SFE's significant improvements in long-term forecasting stem from several complementary and plausible mechanisms.

Contextual disambiguation of ambiguous signals. One of the primary hypothesized roles of SFE is to act as a context provider, breaking the feature symmetry of low-activity nodes. Without SFE, the model attempts to learn a mapping $f(x_i)$ from the input traffic signal. However, after normalization, the inputs from different sleep nodes i and j become nearly indistinguishable, i.e., $x_i(t) \approx x_j(t)$. By augmenting the input, SFE encourages the model to learn a more powerful conditional mapping, $f(x_i, s_i)$. Since the static slack features s_i and s_j are distinct for nodes with different historical activity, the model is provided with the necessary information to learn unique representations. This enhancement of feature discriminability is a crucial first step toward building a more robust and complete understanding of the network.

Learned amplification of weak signal influence. Building on this, the provision of a stable context may enable a form of learned signal amplification. We posit that the model's internal mechanisms can learn to use the static feature s_i as a cue to modulate the importance of the dynamic signal x_i . Conceptually, this can be viewed as the model learning to form an internal representation z_i that is a function of both inputs:

$$z_i = \mathcal{M}(x_i, s_i), \quad (12)$$

where the mapping $\mathcal{M}$ learns to amplify or attenuate the influence of x_i based on the context provided by s_i . This learned, indirect amplification during the forward pass can lead to a more

4.2.4. Why SFE reduces training time (Q2.2)

The reduction in training time when using SFE can be understood from an optimization dynamics perspective. In gradient-based optimization, model parameters w are updated iteratively to minimize a loss function L . The standard update rule is:

$$w_{k+1} = w_k - \eta \frac{\partial L}{\partial w_k}, \quad (13)$$

where η is the learning rate. The convergence speed is thus critically dependent on the quality and magnitude of the gradient $\frac{\partial L}{\partial w_k}$.

For sleep nodes, their low-variance and near-constant signals result in their contribution to the total loss being minimal and relatively insensitive to parameter changes. Consequently, the gradient components associated with sleep nodes (w_s) are often orders of magnitude smaller than those associated with active nodes (w_h):

$$\left\| \frac{\partial L}{\partial w_s} \right\| \ll \left\| \frac{\partial L}{\partial w_h} \right\|. \quad (14)$$

This gradient dominance by active nodes creates a challenging optimization landscape. The overall learning process is slowed down because the optimizer struggles to find a direction that is beneficial for *all* nodes simultaneously, leading to inefficient and prolonged training.

We hypothesize that SFE improves training efficiency by acting as a form of *regularization that improves the conditioning of this optimization problem*. By augmenting the input with a stable and informative slack feature s_i , SFE provides a more consistent signal that helps to balance the learning dynamics. This has two potential benefits:

First, it may smooth the loss landscape, helping the optimizer to find a more direct path to a good minimum. Second, from a numerical optimization standpoint, it may improve the problem's conditioning. The convergence rate is influenced by the condition number κ of the loss function's Hessian matrix H . The number of iterations N to reach a certain accuracy ϵ is approximately:

$$N = O\left(\kappa \ln \frac{1}{\epsilon}\right). \quad (15)$$

By providing a more stable and discriminative input representation, SFE may effectively improve the problem's conditioning (i.e., reduce κ). A better-conditioned problem allows the optimizer to converge more rapidly, leading to a shorter training time.

In summary, SFE improves training efficiency not by naively "boosting" gradients, but by enriching the input space in a way that regularizes the learning process and creates a better-conditioned, more favorable optimization landscape. This allows the model to converge to a robust solution more quickly, which is consistent with the empirical results on training time reduction.

4.2.5. Why the slack feature is computed offline from long-term history (Q3.1)

Rationale. The purpose of the Slack Feature is to capture the *intrinsic* activity level of each node rather than transient traffic shocks. Let $v_{i,t}$ denote the observed speed at node i and time t . We define a long-horizon empirical mean:

$$\mu_i = \frac{1}{H} \sum_{t=t_0}^{t_0+H-1} v_{i,t}, \quad (16)$$

where H spans multiple months (or years). Under the mild assumption that $\{v_{i,t}\}$ is weakly stationary with variance σ_i^2 , the variance of μ_i decays as:

$$\text{Var}[\mu_i] = \frac{\sigma_i^2}{H}, \quad (17)$$

implying that sporadic events such as road construction or accidents (whose duration $L \ll H$) perturb μ_i by at most $O(L/H)$. Consequently, μ_i provides a *statistically stable* estimate of node activity, immune to short-term anomalies. This design ensures that localized events such as temporary road closures, accidents, or construction activities—whose timescales are much shorter than the historical aggregation window—have negligible impact on the Slack Feature, preserving its robustness across real-world conditions.

Offline construction & deployment flexibility. By pre-computing the slack vector offline and freezing it during inference, we decouple the forecasting backbone from real-time fluctuations, yielding faster convergence and consistent behaviour across training and deployment. Whenever the road network undergoes a major structural change (e.g. lane expansion), practitioners can simply regenerate the slack statistics on the updated traffic records without modifying the model architecture, thereby preserving both *flexibility* and *operational simplicity*.

4.2.6. Why slack feature enhancement rather than architectural modification (Q3.2)

Motivation. One might ask why we introduce an *input-level* Slack Feature instead of redesigning internal network layers to highlight dormant nodes. Our choice is driven by two considerations.

(i) *Model-agnostic flexibility.* Feature augmentation leaves the backbone untouched; thus the same slack vector $\mathbf{m} \in \mathbb{R}^N$ can be concatenated or additively injected into any spatial-temporal architecture (GNN, CNN,

Transformer) without altering its complexity, hyper-parameters, or optimisation dynamics. This plug-and-play property is crucial for practical deployment in heterogeneous forecasting pipelines.

(ii) *Normalisation imbalance as the root cause.* Dormant-node under-representation arises primarily from *global feature normalisation* rather than insufficient network capacity. Let $x_{i,t}$ be the raw speed at node i and time t . Standard practice applies a joint z-score transformation across all nodes:

$$\tilde{x}_{i,t} = \frac{x_{i,t} - \mu_{\text{all}}}{\sigma_{\text{all}}}, \quad \mu_{\text{all}} = \frac{1}{N} \sum_{j=1}^N \mu_j, \quad (18)$$

where $\mu_j = \mathbb{E}[x_{j,t}]$ and $\sigma_{\text{all}}^2 = \frac{1}{N} \sum_j \sigma_j^2$. Because high-traffic nodes dominate σ_{all} , the signal-to-noise ratio (SNR) of a low-traffic node k satisfies

$$\text{SNR}_k = \frac{\text{Var}(x_{k,t})}{\sigma_{\text{all}}^2} \ll \frac{\text{Var}(x_{h,t})}{\sigma_{\text{all}}^2}, \quad h = \text{high-traffic node}, \quad (19)$$

which compresses dormant-node gradients and hampers learning even in expressive models.

How SFE resolves the imbalance.

By injecting the long-term slack magnitude as an auxiliary channel, we effectively *re-weight* node contributions prior to any shared normalisation, restoring the relative amplitude of weak signals without disrupting spatial dependencies. Thus SFE targets the true bottleneck—Eqs. (18) and (19)—while retaining full compatibility with diverse forecasting backbones.

4.2.7. Why slack feature enhancement is robust in practical scenarios (Q3.3)

Scenario A: asynchronous or partially missing data.

Let $\mathbf{X}_t = (x_{1,t}, \dots, x_{N,t})^\top$ be the speed vector at time t and $\mathbf{m} = (m_1, \dots, m_N)^\top$ the offline slack vector. Introduce a binary mask $\mathbf{M}_t = \text{diag}(\delta_{1,t}, \dots, \delta_{N,t})$, where $\delta_{i,t} = 1$ if node i is observed at t and 0 otherwise. After applying the embedding strategy in Eq. (5), the model input becomes

$$\tilde{\mathbf{X}}_t = [\mathbf{M}_t \mathbf{X}_t \parallel \underbrace{\mathbf{1} \cdot \mathbf{m}^\top}_{\text{Slack channel}}], \quad (20)$$

where the slack channel is *fully observable* ($\mathbf{1}$ is an all-ones column), even when many $x_{i,t}$ are missing. Thus dormant-node information is consistently supplied to the backbone, preventing the feature space from collapsing when observations are sparse or asynchronous.

Scenario B: predicting traffic outside the current sensor network.

For a new node j^* that lacks historical statistics, one may first collect a short warm-up window of length H^* (e.g. two weeks) and compute $\mu_{j^*} = \frac{1}{H^*} \sum_{t=1}^{H^*} x_{j^*,t}$ followed by the same min-max inversion in Eq. (3). Because Slack Feature generation is an *offline* step, no retraining of the backbone is required; only the slack vector $\mathbf{m}$ is augmented with m_{j^*} . This enables straightforward expansion to unseen regions.

Scenario C: large-scale network updates.

When extensive lane additions or permanent rerouting substantially shift long-term flow patterns, practitioners can simply regenerate the slack statistics on the updated traffic logs and continue inference without changing model parameters, thanks to the decoupled offline design.

4.2.8. What are the differences between the two embedding strategies? (Q3.4)

The Slack Feature can be integrated into the backbone in two ways—*direct concatenation* and *residual embedding*.¹ Both expose long-term node-activity cues, but they differ in parameter cost, computation, and interaction behaviour.

(a) *Direct concatenation.* As defined in Eq. (5), the slack vector is appended as an extra channel, resulting in $\mathbf{x}_{\text{out}} \in \mathbb{R}^{B \times T \times N \times (C+1)}$. Because it introduces *no* new parameters, this option is extremely light-weight; the downstream network autonomously decides how much to rely on the slack signal. Empirically, concatenation excels in smaller backbones where feature independence is preferred.

(b) *Residual embedding.* Residual embedding first projects the slack vector via Eq. (6) and then injects it additively (Eq. (7)). Although this adds a modest number of parameters (W, b) and slightly increases FLOPs, it blends slack information directly into the latent space, often accelerating convergence when the backbone has sufficient capacity.

Practical takeaway. In low-budget settings we recommend direct concatenation for its zero-cost simplicity; for larger models, residual embedding can provide a small but consistent accuracy gain by enriching feature interactions.

5. Experiments and evaluation

Our experimental study is structured around nine core research questions, each formulated to systematically address the key motivations, theoretical insights, and methodological innovations presented in this work. These questions guide our evaluation of the Slack Feature Enhancement (SFE) across diverse tasks and settings:

- **Q1:** How does SFE perform in long-term prediction?
- **Q2:** How does SFE perform in extra-long-term prediction?
- **Q3:** Is SFE effective for short-term prediction as well?
- **Q4:** How robust is SFE? Does it remain effective in enhancing performance under adverse real-world conditions?
- **Q5:** What is the impact of SFE on the model's resource consumption, such as memory usage, inference time and training time?
- **Q6:** Is the SleepNode phenomenon truly pervasive in real-world traffic networks?
- **Q7:** Do the experiments provide strong evidence that SFE genuinely improves traffic flow learning by addressing the SleepNode issue?
- **Q8:** Does SFE truly reflect real-world conditions? In particular, is it reasonable to rely on historical aggregated information for predicting future outcomes?
- **Q9:** Does SFE Improve Model Stability?

5.1. Experimental settings

This subsection consolidates all experimental ingredients required for reproducibility. Section 5.1.1 describes every dataset in detail, Section 5.1.2 lists the competing baselines, and Section 5.1.3 specifies the training protocol and evaluation metrics.

5.1.1. Datasets

We evaluate the performance of our proposed SFE in long-term and extra-long-term forecasting on three real-world datasets: **PEMS-04**, **PEMS-08**, and **Seattle Loop**. For short-term prediction and real-world validation we additionally use **METR-LA** and **PEMS-BAY**. To analyse the prevalence of the SleepNode phenomenon we further incorporate **PEMS-03** and **PEMS-07**. Table 1 summarises key statistics; the full descriptions follow verbatim.

1. **PEMS-03:** Collected from Caltrans District 3 (Sacramento area), PEMS-03 contains traffic flow data from 358 sensors recorded every five minutes between September and November 2018, totaling 26,208 time steps.
2. **PEMS-04:** PEMS-04, from Caltrans District 4 (San Francisco Bay Area), includes data from 307 sensors collected at five-minute intervals from January to February 2018, with 16,992 time steps.

¹ See Eqs. (5)–(7) for the formal definitions.

Table 1
Summary of datasets.

Dataset	# Time Steps	# Nodes	Time Range
PEMS-03	26,208	358	2018/09–2018/11
PEMS-04	16,992	307	2018/01–2018/02
PEMS-07	28,224	883	2017/05–2017/08
PEMS-08	17,856	170	2016/07–2016/08
METR-LA	34,272	207	2012/03–2012/06
PEMS-BAY	52,116	325	2017/01–2017/05
Seattle Loop	8760	323	2015/01–2015/12

3. **PEMS-07**: Sourced from Caltrans District 7 (Los Angeles area), PEMS-07 comprises 883 sensors with data from May to August 2017 at five-minute intervals, totaling 28,224 time steps.
4. **PEMS-08**: PEMS-08, collected from Caltrans District 8 (San Bernardino area), includes data from 170 sensors recorded every five minutes during July and August 2016, yielding 17,856 time steps.
5. **METR-LA**: METR-LA is a traffic speed dataset from Los Angeles County, featuring 207 sensors with data recorded every five minutes from March to June 2012 (34,272 time steps).
6. **PEMS-BAY**: PEMS-BAY contains traffic speed data from 325 sensors in the San Francisco Bay Area, collected every five minutes from January to May 2017, resulting in 52,116 time steps.
7. **Seattle Loop**: The Seattle Loop dataset provides hourly traffic flow data from 323 detectors across the Seattle area, covering January to December 2015 (8,760 time steps).

5.1.2. Baselines

We evaluate the performance of our proposed method against a range of baseline models commonly used in traffic forecasting:

1. **HA [10] (Historical Average)**: A simple yet effective benchmark that predicts future traffic flow by averaging historical data.
2. **VAR [35] (Vector AutoRegression)**: A classical multivariate time series model that captures linear interdependencies across multiple time series.
3. **STGCN [19] (Spatial-Temporal Graph Convolutional Network)**: Employs graph convolutions in combination with temporal convolutions to jointly model spatial and temporal dependencies.
4. **ASTGCN[27] (Attention-based Spatial-Temporal Graph Convolutional Network)**: An extension of STGCN incorporating attention mechanisms to dynamically weight spatial and temporal features.
5. **DCRNN [36] (Diffusion Convolutional Recurrent Neural Network)**: Integrates diffusion convolution operations with Gated Recurrent Units (GRUs) to more effectively capture spatial-temporal dependencies.
6. **GWNet [21] (Graph WaveNet)**: Combines dilated causal convolutions with graph convolutions, adept at modeling dynamic correlations in traffic networks.
7. **GMAN [37] (Graph Multi-Attention Network)**: Utilizes spatial attention mechanisms with vertex grouping to capture complex spatial dependencies.
8. **AGCRN [38] (Adaptive Graph Convolutional Recurrent Network)**: Learns node-specific patterns through adaptive graph structures, obviating the need for a predefined adjacency matrix.
9. **DMSTGCN [22] (Dynamic and Multi-faceted Spatial-Temporal Graph Convolution Network)**: Employs multiple spatial learning strategies and a fusion module for richer feature representations.
10. **SSTBAN [8] (Self-Supervised Spatial-Temporal Bottleneck Attention Network)**: Introduces a bottleneck attention design to achieve computational efficiency in modeling spatial-temporal dependencies.
11. **Extralonger [9]**: A recent state-of-the-art model that unifies spatial and temporal factors through resource-efficient techniques, demonstrating strong performance in both long-term and extra-long-term forecasting scenarios.

5.1.3. Experimental setup and evaluation metrics

We evaluate the effectiveness of our proposed Slack Feature Enhancement (SFE) by integrating it into a range of classic and state-of-the-art traffic forecasting models. To isolate the impact of SFE, we adopt two integration strategies at the input layer—direct concatenation and residual connection—while keeping all other components identical to the original open-source baselines. Importantly, we perform each experiment with multiple independent repetitions to ensure statistical robustness and reliability.

Experimental Environment. All experiments are conducted using PyTorch on a workstation equipped with an NVIDIA RTX 4090D GPU and an Intel Xeon(R) Gold 6430 CPU. We strictly adhere to the original baselines' hyperparameter configurations—batch size, learning rate, optimizer settings, weight decay, etc.—to avoid any confounding factors and ensure a fair comparison. We assess performance over three categories of forecasting horizons: **Short-Term**: Forecasting 6 and 12 time steps. **Long-Term**: Following the settings in Extralonger, we use horizons of 24, 36, and 48 time steps. **Extra-Long-Term**: We consider horizons of 64, 144, 288, 576, 864, 1152, 1440, 1728, and 2016 time steps.

To comprehensively evaluate forecasting accuracy, we adopt four complementary metrics: **Root Mean Square Error (RMSE)** measures the standard deviation of prediction errors and penalizes large deviations; **Mean Absolute Error (MAE)** reports the average magnitude of errors in the same units as the data, offering an intuitive measure of overall accuracy; **Mean Absolute Percentage Error (MAPE)** expresses errors as percentages of true values, enabling scale-invariant comparison across sensors with different traffic volumes; **Weighted MAPE (WMAPE)** further adjusts for aggregate volume by weighting each error term by its ground-truth flow, improving robustness when low-traffic nodes dominate the dataset.

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}, \quad (21a)$$

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|, \quad (21b)$$

$$\text{MAPE} = \frac{1}{N} \sum_{i=1}^N \left| \frac{y_i - \hat{y}_i}{y_i} \right|, \quad (21c)$$

$$\text{WMAPE} = \frac{\sum_{i=1}^N |y_i - \hat{y}_i|}{\sum_{i=1}^N y_i}. \quad (21d)$$

5.2. Results and analysis

5.2.1. Long-term forecasting performance (Q1)

To rigorously evaluate the efficacy of the proposed Slack Feature Enhancement (SFE) in long-term traffic prediction, we integrated SFE with three distinct forecasting frameworks: **DMSTGCN**, **SSTBAN**, and the current state-of-the-art **Extralonger**. These three models are selected due to their recent state-of-the-art performance, representativeness in long-term forecasting literature, and the availability of reliable open-source implementations. The performance metrics are reported in Table 2.

First, integrating SFE with the sub-state-of-the-art baseline, **DMSTGCN**, yielded beneficial results. Specifically, on the **PEMS04** dataset, the DMSTGCN-SFE variant achieved average reductions of **1.14 %** in RMSE, **1.37 %** in MAE, and **1.45 %** in MAPE. On the **PEMS08** dataset, DMSTGCN-SFE improved RMSE by **1.33 %**, MAE by **2.29 %**, and MAPE by **2.02 %**. Finally, on the **Seattle Loop** dataset, DMSTGCN-SFE reduced RMSE by **1.36 %**, MAE by **1.49 %**, and MAPE by **2.08 %**.

Second, when SFE was applied to another sub-state-of-the-art baseline, **SSTBAN**, the performance gains were also noteworthy. On the **PEMS04** dataset, the SSTBAN-SFE variant achieved average reductions of **0.90 %** in RMSE, **0.72 %** in MAE, and **0.90 %** in MAPE. On the **Seattle Loop** dataset, SSTBAN with SFE showed average reductions of **1.97 %** in

Table 2

Performance comparison on PEMS04, PEMS08, and Seattle Loop datasets. **Note:** Bold values indicate the best performance; underlined values denote the second best. SF denotes direct linking of Slack features, while SF* denotes residual linking.

Dataset	Method	24 Time Steps			36 Time Steps			48 Time Steps		
		RMSE	MAE	MAPE	RMSE	MAE	MAPE	RMSE	MAE	MAPE
PEMS04	HA	81.57	56.47	45.49	106.58	76.01	68.84	127.28	93.37	94.62
	DCRNN	42.86	28.70	21.23	51.40	33.78	27.10	57.85	38.26	33.73
	GWNet	35.52	22.79	16.04	38.17	24.71	17.67	40.60	26.42	18.99
	GMAN	38.10	21.67	17.78	52.86	22.12	16.43	47.85	23.35	17.98
	AGCRN	34.44	21.63	14.65	38.19	24.15	16.33	38.26	24.18	16.31
	DMSTGCN	32.09	20.32	14.13	34.86	22.47	15.86	35.05	22.50	16.56
	DMSTGCN + SF	<u>31.77</u>	20.08	13.84	34.28	22.07	15.63	34.78	22.24	16.42
	SSTBAN	33.15	20.36	14.73	34.15	20.82	14.83	35.81	21.66	15.90
	SSTBAN + SF*	32.64	20.17	14.89	33.83	20.61	14.93	35.73	21.61	15.30
	Extralonger	32.01	19.69	13.60	33.52	20.58	14.13	34.41	21.15	14.65
	Extralonger + SF*	<u>31.66</u>	<u>19.47</u>	<u>13.14</u>	<u>32.85</u>	<u>20.14</u>	<u>13.61</u>	<u>33.79</u>	<u>20.86</u>	<u>14.07</u>
	Extralonger + SF	31.52	19.42	13.13	32.90	20.11	13.57	33.71	20.65	14.08
PEMS08	HA	69.72	48.30	32.09	92.72	65.99	46.64	111.85	81.51	61.29
	DCRNN	33.34	22.60	15.46	39.37	25.82	18.53	45.64	30.47	25.10
	GWNet	29.47	19.07	12.25	33.54	21.76	13.68	34.20	22.60	14.16
	GMAN	34.29	17.38	15.66	35.89	17.21	16.33	48.54	18.70	16.81
	AGCRN	28.05	17.45	11.25	30.96	19.39	12.73	31.11	19.46	12.88
	DMSTGCN	26.55	16.75	11.44	28.50	18.15	12.64	28.94	18.34	12.93
	DMSTGCN + SF	26.17	16.37	11.14	28.14	17.71	12.35	28.55	17.92	12.76
	SSTBAN	26.32	15.97	12.29	28.30	16.84	12.20	28.82	16.94	12.47
	Extralonger	26.39	15.89	10.49	28.10	16.72	11.36	29.01	17.31	11.66
	SSTBAN + SF	<u>25.86</u>	15.74	11.45	27.58	16.07	<u>11.22</u>	28.64	16.85	<u>11.24</u>
	Extralonger + SF*	25.91	<u>15.55</u>	<u>10.22</u>	<u>27.55</u>	<u>16.30</u>	<u>11.01</u>	<u>28.33</u>	16.70	11.34
	Extralonger + SF	25.85	15.45	10.16	27.45	16.35	10.92	28.18	<u>16.77</u>	11.22
Seattle Loop	HA	11.86	8.07	26.57	12.37	8.47	27.76	12.31	8.49	27.82
	DCRNN	7.97	4.37	14.04	8.38	4.60	14.41	8.63	4.73	14.91
	GWNet	7.84	4.28	14.06	8.18	4.60	14.92	8.35	4.67	15.04
	GMAN	7.84	4.13	12.88	8.10	4.23	12.95	8.09	4.26	13.26
	AGCRN	7.83	4.27	13.53	8.31	4.64	14.17	8.48	4.75	15.02
	DMSTGCN	7.59	4.08	13.51	7.98	4.31	14.31	8.20	4.49	14.86
	DMSTGCN + SF	7.49	4.01	13.22	7.84	4.26	14.15	8.10	4.41	14.40
	SSTBAN	7.72	4.05	12.69	7.83	4.11	12.44	7.88	4.12	12.35
	SSTBAN + SF	7.52	<u>4.01</u>	12.44	7.68	4.05	12.15	7.77	4.09	12.11
	Extralonger	7.43	4.04	12.48	7.51	4.05	<u>11.96</u>	7.68	4.11	12.04
	Extralonger + SF*	<u>7.21</u>	<u>3.88</u>	<u>12.01</u>	<u>7.36</u>	<u>4.01</u>	11.99	<u>7.54</u>	3.99	<u>11.93</u>
	Extralonger + SF	7.16	3.86	11.88	<u>7.39</u>	3.96	11.95	7.50	<u>4.01</u>	11.89

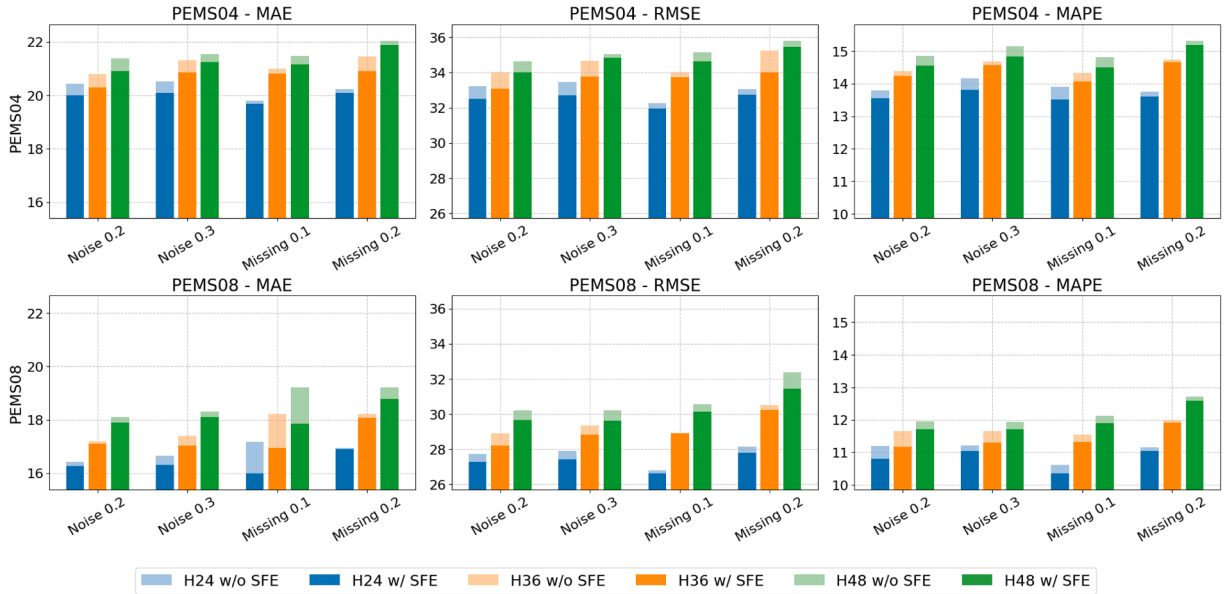


Fig. 4. Robustness evaluation of SFE on PEMS04 and PEMS08 under various adverse real-world conditions (bar chart version). Each row corresponds to a dataset (PEMS04 and PEMS08), while each column corresponds to an evaluation metric (MAE, RMSE, MAPE). For each subplot, we compare the performance of the state-of-the-art model (Extralonger) with and without our proposed SFE under four conditions: moderate noise (Noise 0.2), severe noise (Noise 0.3), moderate missing data (Missing 0.1), and severe missing data (Missing 0.2). Each group of bars corresponds to a specific prediction horizon (24, 36, or 48), where the lighter bar shows the baseline without SFE and the darker, overlapping bar indicates the model equipped with SFE.

Table 3

Performance comparison in extra-long-term scenarios. TS: Length of Time Steps. VAR: Vector AutoRegressive model—a classical multivariate linear forecasting baseline.

Dataset	TS	VAR			Extralonger			Extralonger + SF		
		RMSE	MAE	MAPE	RMSE	MAE	MAPE	RMSE	MAE	MAPE
PEMS04	64	57.14	36.66	35.90	35.11	21.67	14.85	34.43	21.14	14.51
	144	77.22	53.63	64.87	39.33	23.66	16.04	37.90	23.10	15.59
	288	50.43	34.98	34.22	40.46	24.18	16.70	39.95	23.68	16.34
	576	52.90	36.99	35.55	42.15	26.08	19.32	40.25	24.25	16.48
	864	54.55	38.32	37.43	42.98	26.60	19.36	40.42	24.23	16.36
	1152	55.99	39.50	39.12	42.98	26.88	19.33	41.96	24.79	16.43
	1440	56.64	40.05	39.40	43.00	26.62	19.24	42.32	25.41	16.66
	1728	57.01	40.40	39.69	43.33	27.09	19.63	42.70	25.26	16.63
	2016	56.21	39.64	39.23	43.26	27.52	20.21	42.99	25.57	16.88
PEMS08	64	52.01	48.40	36.11	30.01	17.91	12.32	29.64	17.61	11.86
	144	89.00	61.09	46.52	32.17	18.84	12.85	31.85	18.51	12.53
	288	56.07	38.51	26.10	33.76	19.38	13.05	32.39	19.25	12.75
	576	63.16	43.37	28.59	37.73	22.06	14.23	34.40	19.85	13.25
	864	65.07	45.25	30.93	38.57	22.57	15.04	35.36	19.87	13.02
	1152	66.00	45.91	32.42	40.45	23.50	16.03	37.16	20.25	13.38
	1440	67.96	47.03	32.69	41.44	23.73	17.90	37.55	20.60	14.02
	1728	67.36	46.53	32.34	40.85	23.68	18.42	37.23	20.61	14.95
	2016	65.45	44.95	31.37	41.38	23.69	19.18	38.89	20.73	15.82
Seattle Loop	64	11.20	7.39	24.88	7.82	4.34	12.66	7.68	4.13	11.99
	144	11.17	7.33	24.75	8.08	4.31	12.63	7.80	4.15	12.17
	288	11.09	7.31	24.14	8.19	4.41	13.99	7.94	4.25	12.91
	576	11.30	7.43	24.83	8.43	4.64	14.52	8.11	4.40	13.70
	864	11.42	7.49	25.33	8.75	4.71	15.20	8.25	4.41	14.15
	1152	11.53	7.54	25.77	8.15	4.41	13.30	8.09	4.35	13.21
	1440	11.59	7.58	26.01	8.52	4.80	14.43	8.29	4.49	13.79
	1728	11.70	7.65	26.49	8.88	4.95	14.77	8.39	4.66	14.25
	2016	11.76	7.71	26.78	8.18	4.43	13.78	7.74	4.08	12.79

Table 4

Performance comparison of STGCN (the lower the better).

Dataset	Method	6 time steps			12 time steps		
		MAE	RMSE	WMAPE	MAE	RMSE	WMAPE
METR-LA	STGCN	6.51	11.84	0.128	8.21	14.55	0.162
	STGCN + SF	6.40	11.69	0.127	7.70	14.20	0.152
PEMS-BAY	STGCN	2.10	4.168	0.034	2.25	4.70	0.0367
	STGCN + SF	2.04	4.13	0.033	2.20	4.60	0.0357

Table 5

Performance comparison of ASTGCN (the lower the better).

Dataset	Method	6 time steps			12 time steps		
		RMSE	MAE	MAPE	RMSE	MAE	MAPE
PEMS04	ASTGCN	37.46	23.94	16.12	35.31	22.52	15.36
	ASTGCN + SF	35.52	22.57	14.72	33.42	21.05	13.93
PEMS08	ASTGCN	30.74	20.06	12.33	28.91	18.59	11.45
	ASTGCN + SF	28.67	18.68	11.05	27.69	17.98	10.90

RMSE, **1.06 %** in MAE, and **2.08 %** in MAPE. Moreover, on the **PEMS08** dataset, SSTBAN-SFE demonstrated even more pronounced improvements, with reductions of **1.64 %** in RMSE, **2.18 %** in MAE, and **8.24 %** in MAPE.

Finally, on the current state-of-the-art model, **Extralonger**, the incorporation of SFE also resulted in substantial gains. On the **PEMS04** dataset, SFE reduced RMSE by an average of **1.80 %**, MAE by **2.00 %**, and MAPE by **3.77 %**. On the **Seattle Loop** dataset, integrating SFE into Extralonger further lowered RMSE by **2.53 %**, MAE by **3.04 %**, and MAPE by **2.05 %**. Similarly, on the **PEMS08** dataset, SFE yielded average improvements of **2.41 %** in RMSE, **2.70 %** in MAE, and **3.60 %** in MAPE.

Furthermore, to facilitate a more comprehensive understanding of the embedding strategies, we also report the results of integrating SFE using both **direct concatenation** (SF) and **residual connection** (SF*)

on the state-of-the-art Extralonger model. As shown in **Table 2**, both embedding schemes consistently outperform the baseline across all datasets and horizons, with direct concatenation (SF) typically yielding slightly better performance in most cases. These results highlight the robustness and versatility of SFE, regardless of the specific integration form.

We conjecture that the varying degrees of improvement observed across datasets and prediction horizons are closely related to the proportion of persistently low-activity (sleep) nodes and the severity of information sparsity in each setting. In particular, datasets like **PEMS08** and longer forecasting horizons typically feature a higher fraction of dormant nodes and weaker temporal correlations, making it much harder for baseline models to **maintain a complete and unbiased network representation**. SFE explicitly injects node-specific historical context, which compensates for this loss of dynamic information from sleep nodes and becomes increasingly beneficial as the prediction horizon

grows. This effect is most pronounced in challenging scenarios—such as extra-long-term forecasting or datasets with many low-activity nodes—where SFE enables the model to maintain predictive accuracy even as traditional temporal signals fade. Conversely, in settings with fewer sleep nodes or shorter horizons, the baseline model can already capture most temporal structure, so the incremental gains from SFE are less marked.

Overall, these results substantiate that augmenting a forecasting model with the proposed Slack Feature—as an auxiliary spatiotemporal signal—enables it to capture more nuanced traffic dynamics, thereby yielding significantly more accurate long-term predictions.

5.2.2. Extra-long-term prediction performance (Q2)

To assess the efficacy of the Slack Feature Enhancement (SFE) in the context of extra-long-term traffic forecasting, we conducted experiments using the state-of-the-art Extralonger model, which is currently the only model capable of extra-long-term predictions and represents the SOTA in this domain. The experimental results demonstrate that, on the PEMS04 dataset, the integration of SFE yielded an average reduction of 2.41 % in RMSE, 5.59 % in MAE, and 11.60 % in MAPE. On the Seattle Loop dataset, SFE achieved average reductions of 3.61 % in RMSE, 5.07 % in MAE, and 5.05 % in MAPE. On the PEMS08 dataset, the improvements were even more pronounced, with average reductions of 6.23 % in RMSE, 9.28 % in MAE, and 12.55 % in MAPE. Detailed performance comparisons are presented in Table 3.

Furthermore, when the prediction horizon is extended further—specifically, when forecasting beyond 1152 time steps (corresponding to a prediction horizon of four days or more) on the PEMS08 dataset—the performance gains become significantly more substantial, with average reductions of 8.10 % in RMSE, 13.17 % in MAE, and 18.64 % in MAPE.

These findings substantiate that the incorporation of SFE greatly enhances the model's ability to capture complex spatiotemporal traffic features over extra-long horizons. While short-term predictions benefit from strong temporal correlations among adjacent time steps, long-term forecasts require a nuanced understanding of historical traffic dynamics, as traffic conditions in one period are not necessarily indicative of those in a subsequent period. By directly leveraging comprehensive historical traffic information to generate enriched feature representations, our SFE approach enables the model to learn these intricate dependencies, thereby delivering markedly improved forecasting performance as the prediction horizon increases.

We conjecture that the varying degrees of improvement observed across datasets and prediction horizons are closely related to the proportion of persistently low-activity (sleep) nodes and the severity of information sparsity in each setting. In particular, datasets like PEMS08 and longer forecasting horizons typically feature a higher fraction of dormant nodes and weaker temporal correlations, making it much harder for baseline models to **maintain a complete and unbiased network representation**. SFE explicitly injects node-specific historical context, which compensates for this loss of dynamic information from sleep nodes and becomes increasingly beneficial as the prediction horizon grows. This effect is most pronounced in challenging scenarios—such as extra-long-term forecasting or datasets with many low-activity nodes—where SFE enables the model to maintain predictive accuracy even as traditional temporal signals fade. Conversely, in settings with fewer sleep nodes or shorter horizons, the baseline model can already capture most temporal structure, so the incremental gains from SFE are less marked.

5.2.3. Short-term prediction performance (Q3)

Although this work primarily focuses on long-term traffic forecasting, we also explored the potential applicability of SFE in short-term settings. To this end, we integrated the proposed Slack Feature Enhancement (SFE) into two classical baseline models, STGCN and ASTGCN, which remain widely adopted in traffic forecasting research due to their

influence and numerous extensions. Experimental results are summarized in Tables 4 and 5.

On the PEMS04 dataset, SFE yields an average reduction of 5.18 % in RMSE, 5.72 % in MAE, and 8.68 % in MAPE when applied to ASTGCN for a 6-time-step horizon, and reductions of 5.35 %, 6.53 %, and 9.31 % for a 12-step horizon. On the METR-LA dataset, integrating SFE with STGCN results in improvements of 1.27 % in RMSE, 1.69 % in MAE, and 0.78 % in WMAPE for 6 steps, and enhancements of 2.41 %, 6.21 %, and 6.17 % for 12 steps.

While SFE was not specifically designed for short-term forecasting, these results suggest that the method may still offer benefits in such scenarios, and highlight its potential as a generally applicable enhancement module.

5.2.4. Evaluation of SFE robustness under adverse real-world conditions (Q4)

In real-world transportation networks, traffic sensors frequently face harsh environmental conditions, such as severe weather, sensor malfunctions, or communication disruptions. These adverse conditions can introduce significant noise and missing data into traffic flow measurements, challenging the robustness and reliability of prediction models. In this subsection, we comprehensively evaluate the robustness of our proposed Slack Feature Enhancement (SFE), aiming to verify its effectiveness under realistic adverse scenarios.

Specifically, we conducted extensive experiments utilizing a state-of-the-art model, Extralonger, across two widely-used benchmark datasets, PEMS04 and PEMS08. Predictions were performed for short-term (24-step), mid-term (36-step), and long-term (48-step) horizons. To simulate realistic adverse scenarios, we designed robustness experiments focusing on two types of challenges:

- **Data Missingness:** We introduced random missing data at rates of 10 % and 20 %, representing moderate and severe missing data scenarios, respectively.
- **Data Noise:** Gaussian noise with standard deviations of 0.2 and 0.3 was added, corresponding to moderate and severe noise conditions.

As illustrated in Fig. 4, the experimental results demonstrate that incorporating SFE consistently improves the performance of the baseline Extralonger model across all evaluation metrics (MAE, RMSE, MAPE), datasets (PEMS04, PEMS08), and prediction horizons (24, 36, and 48 steps), under both noise and missing data scenarios. These outcomes strongly suggest that SFE effectively enhances the robustness and reliability of traffic flow prediction models in adverse real-world conditions.

5.2.5. Resource consumption analysis (Q5)

To rigorously assess the computational overhead associated with the integration of Slack Feature Enhancement (SFE), we examine its impact on memory consumption, inference latency, and training duration.

Memory usage. The SFE is incorporated as an additional feature input without altering the core architecture or substantially increasing the model parameters. Consequently, the overall memory footprint of the SFE-augmented model is nearly identical to that of the baseline model, thereby ensuring minimal impact on memory usage.

Inference time. SFE functions solely as an extra embedding that enriches the feature space without necessitating any significant modifications to the network's computational graph. As a result, the inference latency remains largely unaffected, and the augmented model exhibits inference times comparable to the original model.

Training time. Our preliminary experiments demonstrate that SFE considerably reduces training time. As illustrated in Fig. 5, the average training duration is reduced relative to each model's original training

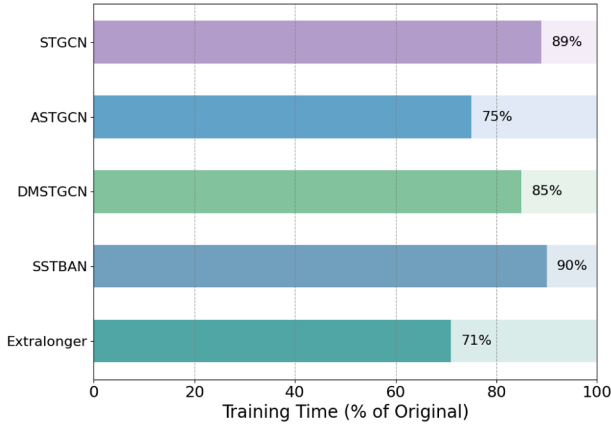


Fig. 5. Comparison of training time reductions across different models after incorporating Slack Feature Enhancement (SFE). The percentages indicate the relative training duration compared to each model's original training time without SFE, based on actual elapsed time (in seconds).

time measured in actual elapsed seconds. Specifically, the average training time was reduced to 71 % for Extralonger, 90 % for SSTBAN, 85 % for DMSTGCN, 75 % for ASTGCN, and 89 % for STGCN.

The reported training time for each method corresponds to the total wall-clock time from the start of training until automatic termination by the backbone model's built-in early stopping mechanism. All experiments used the default early stopping settings provided in the official open-source implementations, where training is stopped once the validation loss fails to improve for a fixed number of epochs.

5.2.6. Empirical analysis of SleepNode phenomenon in real-world traffic networks (Q6)

To investigate the prevalence of the SleepNode phenomenon in practical scenarios, we conducted experiments on four widely used real-world datasets: PEMS03, PEMS04, PEMS07, and PEMS08. In our study, a node is classified as a SleepNode if its average traffic flow falls below 40 % of the global mean. Furthermore, if the traffic flow within a given time period is below approximately 6.6 % of the global mean, we consider that period to be in a zero-flow state.

The experimental results, as illustrated in Fig. 6, reveal that SleepNodes are markedly present across all examined datasets. Specifically, 64 SleepNodes were identified in PEMS03, corresponding to approximately 17.88 % of the total nodes. In PEMS04, 18 SleepNodes were detected, constituting roughly 5.86 % of nodes. Similarly, in PEMS07, 45 nodes (accounting for 5.10 % of the total) met the criteria, while PEMS08 exhibited 17 SleepNodes, representing about 10 % of its nodes. Notably, in the PEMS07 and PEMS08 datasets some SleepNodes demonstrated zero-flow ratios as high as 80 %, suggesting extended periods of negligible traffic.

Collectively, these empirical observations substantiate the widespread occurrence of the SleepNode phenomenon in real-world traffic networks. This evidence is consistent with the intuitive understanding of traffic distribution, whereby significant aggregation often results in certain nodes persistently exhibiting low activity levels.

5.2.7. Empirical evidence of SFE addressing SleepNode issue (Q7)

Deep learning method, by their nature, operate largely as "black boxes," making it challenging to directly and rigorously validate specific mechanisms within the method. Despite this inherent limitation, it remains possible and valuable to provide supporting empirical evidence from indirect perspectives that can substantiate the effectiveness of our proposed method. To quantitatively evaluate the severity of the SleepNode phenomenon discussed in Question 6 (Q6) and previously

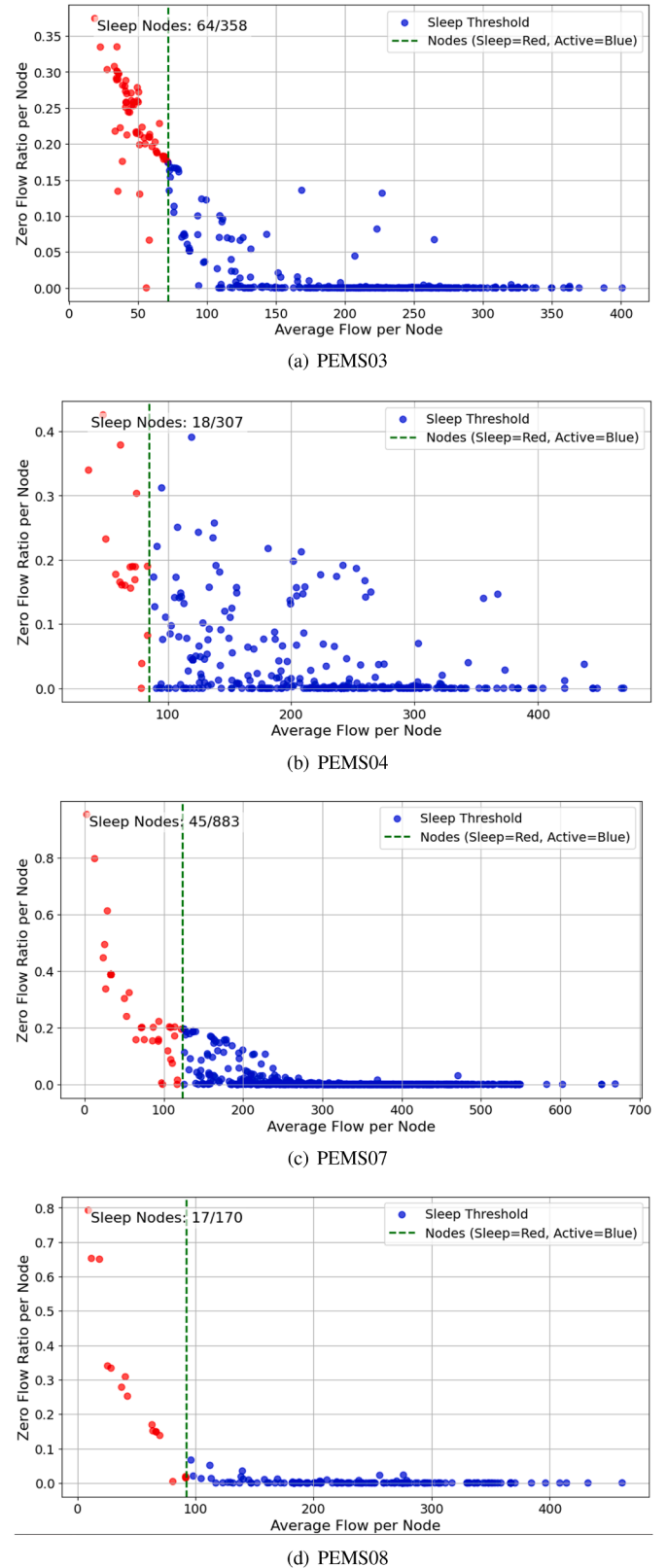


Fig. 6. Visualization of SleepNode phenomena across four real-world datasets. Nodes with an average flow below 40 % of the overall mean are defined as SleepNodes (red), while others are marked as active (blue). The green dashed line represents the threshold separating SleepNodes from active nodes.

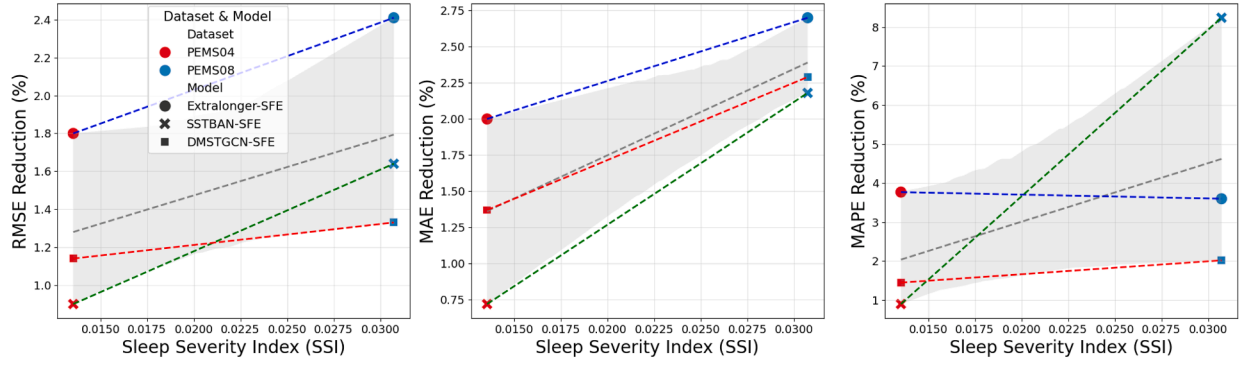


Fig. 7. Relationship between the Sleep Severity Index (SSI) and the reduction percentages of RMSE, MAE, and MAPE achieved by incorporating Slack Feature Enhancement (SFE). Each subplot presents the performance of different models across two datasets, with individual regression lines illustrating model-specific trends. The shaded gray area indicates the overall trend, which shows a positive correlation between higher SSI and larger performance improvements.

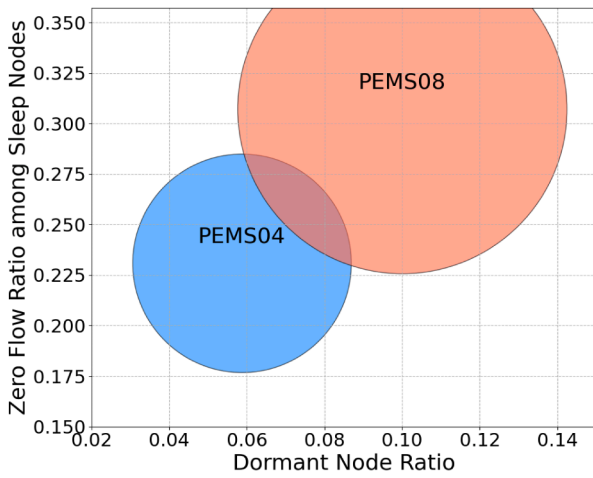


Fig. 8. Bubble chart visualization illustrating Sleep Severity Index (SSI) across different datasets (PEMS04, PEMS08). Larger bubbles indicate greater severity of the SleepNode phenomenon.

illustrated in Fig. 6, we define a metric called the Sleep Severity Index (SSI). Specifically, SSI is calculated as the product of two key indicators: the *Zero Flow Ratio*, representing how frequently nodes exhibit near-zero traffic flow, and the *Sleep Node Ratio*, measuring the proportion of nodes classified as Sleep. Notably, these two metrics remain consistent with definitions and presentations introduced previously in Q6 and Fig. 6.

$$\text{SSI} = \text{Zero Flow Ratio} \times \text{Sleep Node Ratio} \quad (22)$$

In order to empirically verify whether our proposed Slack Feature Enhancement (SFE) approach effectively mitigates the detrimental impact of the SleepNode issue, we hypothesize that datasets exhibiting a higher severity of SleepNode conditions (higher SSI values) should intuitively demonstrate greater performance improvements when incorporating SFE. Hence, we specifically select the *PEMS04* and *PEMS08* datasets as experimental benchmarks due to their pronounced difference in SSI values.

Fig. 8 presents a bubble chart visualization of SSI values across these datasets, where the bubble sizes are directly proportional to the magnitude of the SSI metric. Clearly visible from the figure, the SSI for *PEMS08* substantially exceeds that of *PEMS04*, providing clear evidence of a more pronounced SleepNode phenomenon within the *PEMS08* dataset.

To further explore the hypothesized relationship between SSI and the efficacy of our SFE method, we examine the extent of reduction in prediction errors (RMSE, MAE, and MAPE) resulting from the integra-

Table 6

Ablation study on PEMS04, 36-step horizon. Percentage values denote the average relative change compared to the base model. The "+ LM" setting denotes a variant where our original prior features (Slack Feature) were replaced with randomly initialized, trainable features of the same dimension.

Setting	RMSE	MAE	MAPE	Change (%)
SSTBAN	34.15	20.82	14.83	—
SSTBAN + LM	34.35	21.01	14.79	+0.41 %
Extralonger	33.52	20.58	14.13	—
Extralonger + LM	33.45	20.65	14.07	−0.10 %

tion of SFE into state-of-the-art traffic prediction models. Fig. 7 provides compelling visual evidence of this relationship, clearly indicating a pronounced positive correlation across all three evaluation metrics with increasing SSI. These empirical observations strongly support our hypothesis that datasets with more severe SleepNode phenomena (indicated by higher SSI values) indeed benefit more significantly from the introduction of SFE Table 6.

To verify that the model's performance improvement is indeed due to the proposed prior information, we conducted an ablation study. Specifically, we replaced our original prior features (Slack Feature) with randomly initialized trainable features. The experimental results show that after replacing the prior information, the performance gains were negligible, and in some cases, even led to a slight degradation. This clearly demonstrates that the additional prior information we provided is the key factor contributing to the model's performance enhancement. Collectively, these experimental findings robustly confirm our initial assertion that SFE effectively mitigates the negative impact of the SleepNode phenomenon. Furthermore, the observed performance improvements become particularly pronounced in environments characterized by more severe SleepNode conditions, validating the strategic capability of our proposed model to dynamically emphasize critical spatial-temporal patterns in urban traffic networks.

We also performed a granularity analysis on PEMS04 by ranking nodes by their average traffic and grouping them into low-activity (bottom 20%), medium-activity (middle 60%), and high-activity (top 20%) categories. As shown in Fig. 11, SFE yields the most significant improvements for the two extreme groups: the low-activity and high-activity nodes. This result strongly aligns with our theory. The low-activity nodes (our 'sleep nodes') benefit because SFE provides the necessary context to disambiguate their compressed signals. The high-activity nodes benefit because their low Slack Feature value acts as a powerful and stable identifier, helping the model distinguish these critical network backbones from moderately active nodes. By enhancing the discriminability

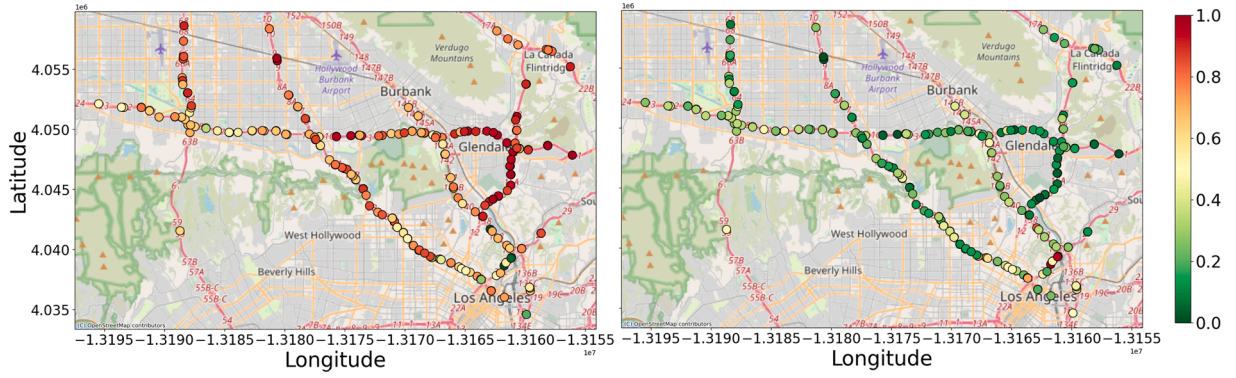


Fig. 9. Comparison between the METR-LA Traffic Flow Heatmap (left), generated from actual traffic flow data during a specific period, and the Slack Heatmap (right), constructed using historical traffic data from the preceding period.

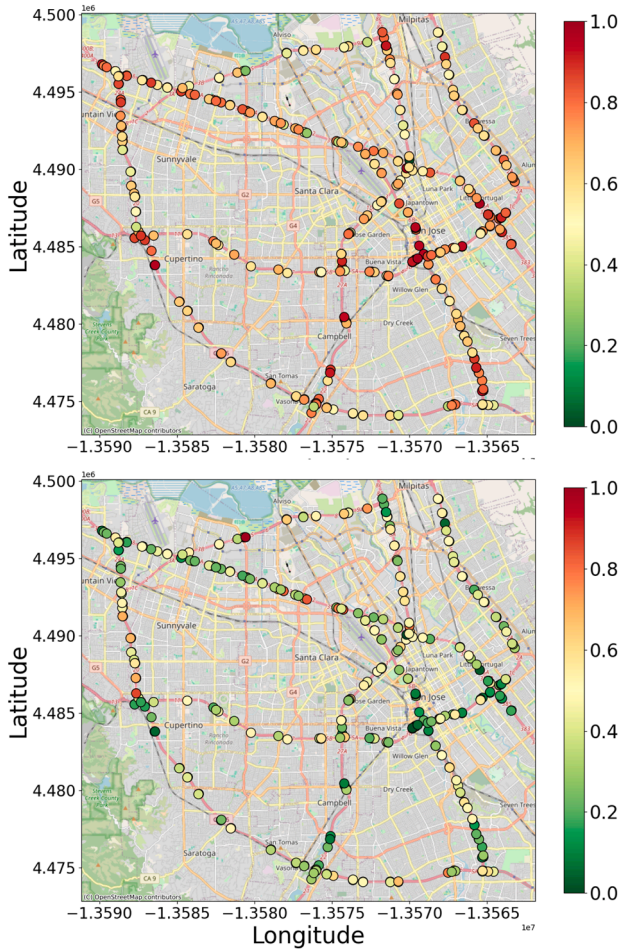


Fig. 10. Comparison between the PEMS-BAY Traffic Flow Heatmap (top), generated from actual traffic flow data during a specific period, and the Slack Heatmap (bottom), constructed using historical traffic data from the preceding period.

at both ends of the activity spectrum, SFE enables the model to learn a more nuanced and accurate representation of the global traffic patterns.

5.2.8. Real-world validity (Q8)

To validate the performance of SFE in real-world scenarios and assess the reasonableness of its underlying premise, we compare two heatmaps. Our evaluation is conducted on two real-world datasets, PEMS-BAY and METR-LA.

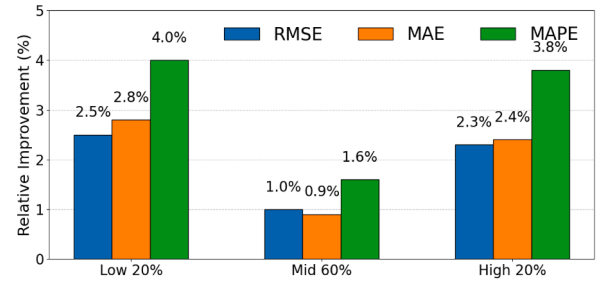


Fig. 11. Relative gain (%) of the Extralonger with SFE.

Fig. 9 shows a comparison between the Traffic Flow Heatmap (left), generated from actual traffic flow data of a later period (representing "future" traffic conditions), and the Slack Heatmap (right), constructed from historical traffic flow data of an earlier period (reflecting past node inactivity levels).

Notably, the slack heatmap clearly highlights high-traffic corridors—depicted in green—such as the Hollywood, Ventura, and Glendale highways, which correspond closely to the high-flow (i.e., high-temperature) regions in the real-world heatmap. Furthermore, segments along Highway 110 that appear as "sleep nodes" (highlighted in red on the slack heatmap) similarly exhibit extremely low traffic flow in the real-world heatmap.

This observation reinforces an important and intuitive argument: although individual nodes experience significant temporal fluctuations in traffic flow, those with historically low traffic tend to remain low-flow relative to other nodes. In the absence of substantial changes in the road network or urban infrastructure, it is highly unlikely for a node with low past traffic to suddenly exhibit a dramatic surge in flow. This finding not only corroborates the validity of using historical aggregated information to forecast future traffic patterns, but also forms one of the key premises underlying our method. In addition to the METR-LA findings, our evaluation on the PEMS-BAY dataset yields analogous results. As depicted in **Figs. 10**, the heatmap generated from the future 30% time period and the slack level map derived from the past 70% of historical data both clearly delineate high-traffic corridors and identify regions that function as sleep nodes. Similar to the METR-LA analysis, these PEMS-BAY results further corroborate the notion that nodes with consistently low historical activity remain in a low-flow state, barring significant alterations in the road network or urban infrastructure. This consistency across datasets reinforces the validity of employing historical aggregated traffic patterns to forecast future conditions, and it substantiates the efficacy of our proposed Slack Feature Enhancement (SFE) method in mitigating the impact of sleep nodes on long-term and extremely long-term traffic forecasting.

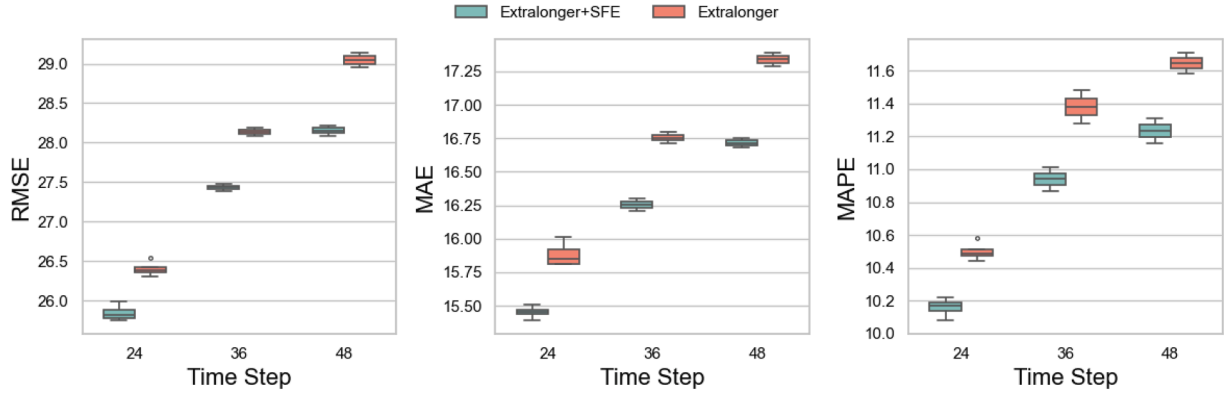


Fig. 12. Stability comparison between Extralonger and Extralonger + SFE across different prediction horizons (24, 36, 48) on the PEMS08 dataset.

5.2.9. Stability analysis of SFE (Q9)

Although all experimental results in this work are based on multiple repeated runs to ensure statistical reliability, we further conduct a detailed stability analysis to provide a more intuitive illustration of the consistency and robustness brought by SFE. Specifically, we select the state-of-the-art baseline Extralonger and focus on the PEMS08 dataset, which exhibits the most severe SleepNode phenomenon. We visualize the results using box plots to depict the distribution and variance of RMSE, MAE, and MAPE metrics under different prediction horizons (24, 36, and 48 time steps).

As shown in Fig. 12, the SFE-augmented models consistently exhibit reduced variance across all three metrics. In addition, we observe that SFE not only stabilizes the additional feature signals but, in most cases, also leads to an overall improvement in the model's intrinsic stability. These results suggest that SFE contributes to more robust learning against the adverse effects of SleepNodes, while maintaining statistical consistency across repeated runs.

6. Conclusion

In this work, we identified a long-standing and often overlooked issue in traffic flow forecasting: the presence of *sleep nodes*, which consistently exhibit low or zero traffic. After standard data preprocessing, these nodes suffer from a significant loss of dynamic information, thereby hindering the effective learning of global traffic patterns and degrading overall generalization performance.

To address this problem, we introduced a simple plug-and-play method termed *Slack Feature Enhancement* (SFE). By leveraging historical data, SFE computes a Slack Level for each node—an inverse measure of its average activity—and fuses this information with real-time inputs at the network's input layer. This approach effectively compensates for the lack of informative signals from sleep nodes and enhances feature discrimination between different types of nodes.

Through extensive experiments combined with a discussion of its plausible mechanisms, we demonstrated that SFE significantly alleviates the detrimental impact of sleep nodes. In long-term and, especially, extremely long-term forecasting scenarios, our method exhibits notable improvements in both training efficiency and predictive performance.

Overall, our study reveals an important yet under-explored issue in long-term traffic forecasting and proposes a viable solution that not only enhances performance in ITS applications but may also offer insightful directions for other time series prediction tasks characterized by highly heterogeneous data. Future research could further refine this approach and explore its broader applicability.

Future work could explore a broader range of feature fusion strategies to further unlock the potential of the proposed SFE framework. For instance, incorporating cross-attention mechanisms might facilitate more dynamic interactions between slack features and real-time

inputs, allowing for more contextually informed integration. Additionally, gated mechanisms may offer adaptive control over the contributions of different feature streams, thereby enhancing the flexibility of the fusion process. Techniques such as conditional normalization and FiLM (Feature-wise Linear Modulation) also represent promising avenues for achieving context-aware feature blending. Evaluating these alternative approaches could further exploit the capabilities of SFE, potentially leading to enhanced feature discrimination and superior forecasting performance.

CRediT authorship contribution statement

Peng Wang: Writing – review & editing, Validation, Supervision, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization; **Hairui Sun:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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